

Minseo Kim

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RESEARCH INTERESTS

Electrochemical energy materials, with a primary focus on rechargeable battery cathodes; layered oxide cathodes for Na-ion and Li-ion batteries; lattice oxygen redox, voltage hysteresis, and interfacial degradation at high voltage; electrolyte/cathode interphase design for stable battery operation.

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** Mar. 2023 – Expected Feb. 2028
B.S. in Materials Science and Engineering; Minor in Chemical and Biomolecular Engineering
◦ Overall GPA: 4.03/4.3; Major GPA: 4.14/4.3
Daejeon, South Korea
- **Technical University of Munich (TUM)** Mar. 2025 – Aug. 2025
Exchange Student, Department of Chemistry
◦ Coursework included electrochemistry, battery storage, fuel cells, and deep learning.
Munich, Germany
- **Hansung Science High School** Mar. 2020 – Feb. 2023
High School Diploma
◦ Project Research Program, Best Paper Award (Jul. 2021)
◦ KAIST President's Doctoral Award (Feb. 2023)
Seoul, South Korea

RESEARCH EXPERIENCE

- **Computational Materials Design and Discovery (CMDD) Lab, KAIST** Dec. 2025 – Present
Undergraduate Research Assistant (PI: Prof. Donghwa Seo)
◦ Conducting research on oxygen-redox-based layered oxide cathodes for rechargeable Na-ion batteries, focusing on P2-type $\text{Na}_{0.6}\text{Li}_{0.2}\text{Mn}_{0.8}\text{O}_2$ (NLMO).
◦ Investigating how high-voltage interfacial side reactions affect the reversibility of lattice oxygen redox, voltage hysteresis, and plateau decay.
◦ Designed charged-state rest experiments with and without electrolyte contact to distinguish electrolyte-driven degradation from bulk structural relaxation.
◦ Performing cathode powder synthesis, electrode fabrication, Na half-cell assembly, galvanostatic cycling, dQ/dV analysis, and electrolyte/additive screening.
◦ Analyzing structural and electrochemical degradation using XRD and electrochemical measurements; preparing further characterization using Raman, XAS, and electron microscopy.
Daejeon, South Korea
- **Energy Conversion & Storage Materials Lab, KAIST** 2.5 months
Undergraduate Research Student, Individual Study (PI: Prof. Eunae Cho)
◦ Synthesized amorphous ruthenium oxide-based nanosheet catalysts for oxygen evolution reaction (OER) in water electrolysis.
◦ Evaluated electrocatalytic activity using RDE-based electrochemical analysis, including polarization curves, Tafel analysis, ECSA calculation, EIS, and stability tests.
Daejeon, South Korea
- **Lab. for Energy and Sustainability, KAIST** 2.5 months
Undergraduate Research Student, Individual Study (PI: Prof. Jihun Oh)
◦ Conducted research on electrochemical CO_2 conversion using Ni single-atom catalyst-coated gas diffusion electrodes.
◦ Performed zero-gap electrolyzer measurements with various membrane configurations.
Daejeon, South Korea

PUBLICATIONS

- [1] M. Kim, Y. Park, D. Seo*, **Effect of Interfacial Side Reactions on the Reversibility of Lattice Oxygen Redox in P2-Type $\text{Na}_{0.6}\text{Li}_{0.2}\text{Mn}_{0.8}\text{O}_2$ Cathodes.** Manuscript in preparation.

ACADEMIC PRESENTATIONS & CONFERENCES

- **Korean Electrochemical Society Spring Meeting** Spring 2026
Conference Attendance
- **ENGE Fall Poster Presentation** Fall 2026
Poster Presentation, title to be determined

RELEVANT COURSEWORK

- MS354 Electrochemistry for Materials Science (A0)
- Applied Electrochemistry (TUM)
- Battery Storage (TUM)
- Fuel Cells in Energy Technology (TUM)
- Introduction to Computational Materials Science (A+)
- CBE301 Chemical and Biomolecular Engineering Laboratory (A+)

TECHNICAL SKILLS

- **Synthesis & Fabrication:** Solid-state synthesis, electrode fabrication (slurry process), electrolyte preparation, coin cell assembly, gas diffusion electrode fabrication
- **Electrochemical Analysis:** Galvanostatic cycling, CV/LSV, RDE, EIS, GITT, dQ/dV, Tafel analysis, ECSA calculation, zero-gap electrolyzer measurements
- **Materials Characterization:** XRD, SEM, TEM, Raman, XPS, ICP-OES
- **Software & Computational Tools:** Origin, Python, L^AT_EX, Athena, FullProf, basic MD/DFT analysis from computational materials coursework

TEACHING EXPERIENCE

- **Language Buddy Program** Spring 2026
KAIST
 - Supported language exchange and communication practice for international students through the KAIST Language Buddy Program.
- **CBE205 Chemical and Biomolecular Engineering Analysis Mentor** Spring 2026
KAIST
 - Mentored international students in CBE205 through concept review and problem-solving support.

HONORS & SCHOLARSHIPS

- **Dean's List** Spring 2026
College of Engineering, KAIST
- **Kim Young-han Global Leader Scholarship** Aug. 2025 – Jul. 2026
KAIST
- **National Science & Technology Scholarship** Mar. 2023 – Sep. 2026
Korea Student Aid Foundation
- **KAIST President's Doctoral Award** Feb. 2023
Hansung Science High School
- **Project Research Program, Best Paper Award** Jul. 2021
Hansung Science High School